MILITARY SPECIFICATION

ALTIMETER AAU-31/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a counter-drum-pointer, integrally lighted, self-contained altimeter (indicator) having a range of -1,000 to 50,000 ft.

1.2 Classification. Two types of altimeters are covered by this specification and differ only in respect to variation in color of the integral white lighting as follows:

I. Type AAU-31/A-1 Altimeter (interchangeable with AAU-31/A) - Incorporates "aviation red" lighting and "blue filtered" white lighting.

II. Type AAU-31/A-2 Altimeter (interchangeable with AAU-31A/A) - Incorporates "aviation red" lighting and "instrument and panel" white lighting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC-6610

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified in the contract or purchase order, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft; General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum Alloys and Aluminum
DOD-D-1000/1	Drawings, Engineering and Associated Data
MIL-C-14806	Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges
MIL-T-18303	Test Procedures, Production, Acceptance and Life for Aircraft Electronic Equipments; Format for
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment; General Requirements for
MIL-L-25467	Lighting, Integral, Aircraft Instrument; General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-L-27160	Lighting, Instrument, Integral, White; General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby Synthetic
MIL-P-21600	Paint System, Fluorescent, for Aircraft Application
MIL-G-81704	Glass, Aircraft Instruments, Lighting Wedges and Cover

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STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards; Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MS-3113	Connector, Electric Solder Type Solder Mounting, Bayonet Coupling
MS-3114	Connectors, Receptacle, Electric Solder Type, Jam Nut Mounting, Bayonet Coupling
MS-33549	Cases, Aircraft Instrument 2 3/4 Inch Dial With Sump, Standard Dimensions for
MS-33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS-33649	Bosses, Fluid Connection, Internal Straight Thread

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

Department of Defense Standardization

SD-6 Provisions Governing Qualification

(Copies available from: Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The general requirements for qualification are specified in Publication SD-6, Provisions Governing Qualification. The production instruments furnished under the specification shall represent a product which has been tested and meets the qualification requirements specified herein under Section 4, Quality Assurance Provisions, and has been listed on or approved for listing on the applicable Qualified Products List.

3.1.1 Retention of qualification. Retention of qualification consists of periodic verification of compliance of the product with the requirements of this specification. The manufacturer shall provide a letter of certification of compliance to the activity responsible for the Qualified Products List at intervals of not more than two years. A copy shall also be submitted to the Naval Air Systems Command, Attention Avionics Division. The letter shall attest that the qualified altimeter is still available from the listed plant and can be produced in the same plant under the same conditions as when qualified. Any changes in parts, materials, design, or processes for manufacture of the altimeter shall be reported as required by SD-6, Provisions Governing Qualification.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.3 Standard general requirements of MIL-E-5400 Class 2 shall apply.

3.3.1 Non-magnetic materials. Non-magnetic materials shall be used for all parts except where magnetic materials are essential. The extent to which magnetic materials are required shall be reported to the procuring activity.

3.3.2 Protective treatment. When materials are used in the construction of the altimeter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of the specification.

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The use of any protective coating that will crack, chip or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.3 Toxic and corrosive fumes. The materials, as installed in the altimeter and under the conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.3.4 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, case inserts, and fine pitch gear teeth need not be anodized. Parts which do not anodize satisfactorily may be coated in accordance with MIL-C-5541.

3.3.5 Components. The electronic and mechanical components shall be in accordance with MIL-E-5400.

3.3.6 Jewel bearings. All sapphire and ruby jewel bearings used in the altimeter shall be in accordance with MIL-B-27497.

3.3.7 Threads. The threads of all machine screws shall conform to MIL-S-7742.

3.4 Design and construction.

3.4.1 Features. The altimeter shall include the following features: a precision all-mechanical altimeter mechanism, a counter-drum-pointer type altitude display, integral red and white lighting, and a barometric setting displayed by means of a four digit counter. The electrical power to operate the integral vibrator is 28 VDC from the emergency power bus. The design of the altimeter should provide for sustained operation under any of the preceding power conditions. The altimeter shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations and such other conditions as are incident to service, shipping, storage and installation without failure.

3.4.2 Maintainability. The maintenance concept for this equipment shall be repair at depot level only.

3.4.2.1 Repair time. The altimeter shall be completely repairable at depot level. The maintenance time shall not exceed 8 hours utilizing one depot technician and authorized depot AGE and technical information. Spare parts and facilities are assumed to be available. Maintenance time shall include all repair and test time excluding government or manufacturer acceptance testing.

3.4.3 Mechanism adjustment. The mechanism shall be provided with a means of adjusting or correcting the indications to conform to the performance required herein, if this should become necessary during the life of the altimeter. The calibration adjustment shall not affect the temperature performance of the instrument. This means of adjustment shall be reasonably simple to manipulate and of such nature that adjustment (rezeroing) can be accomplished by tools ordinarily used by an instrument repairman. The adjustment shall be made in accordance with the manufacturer's instructions (Government approved) at the

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level of maintenance for which such an adjustment is authorized by the government.

3.4.4 Barometric setting system. A setting system, controlled by a knob, shall be provided that will permit the altimeter display to be set to indicate zero altitude at any existing ground level pressure throughout the range of 28.10 to 31.00 (inclusive) inches of mercury in 0.01 inch increments. The gear ratio between the setting knob and the display pointer shall be such that one complete revolution of the setting knob shall result in a display pointer movement of approximately 200 to 300 ft. Clockwise rotation of the setting knob shall result in clockwise rotation of the pointer. The shaft of the setting knob shall be capable of withstanding a load of 15 pounds applied perpendicular to the shaft at the extreme outward end of the knob without permanent deformation. A positive stop or stops shall be provided to limit the barometric setting mechanism movement such that the setting range of 28.0 to 31.1 inches of mercury is not greatly exceeded. The setting system shall not introduce errors greater than specification tolerances when set at any point within the range established by the stops. The stops shall be capable of withstanding a torque of 6 pound inches applied to the barometric setting knob without damage to the stops or the mechanism. If the design is such that the stops or the mechanism may be damaged by the application of 6 pound inches torque to the knob, the use of a torque limiting mechanism incorporated in the knob or in the instrument is acceptable if specification requirements are otherwise met. The design of such a device is subject to the approval of the qualifying agency. The resistance (torque) when setting the barometric pressure shall be within the range of 3 to 10 ounce inches. The barometric counter unit displaying the setting shall be located as shown in Figure 1. The mechanism shall be capable of withstanding a push and a pull of 15 pounds while the barometric scale is rotated through the range of settings from the minimum stop to the maximum stop without disengagement or damage of the mechanism. The barometric setting system shall be designed to withstand a 2 g vibration throughout the frequency range of 5 to 2000 Hz without barometric counter display shift exceeding .01 in. Hg.

3.4.4.1 Adjustment of barometric scale counter to pointer. The barometric setting system, controlled by the setting knob in its extended position, shall permit adjustment of the relative setting of the pointer and barometric scale counter. Unlocking of the zero set locking screw as shown in Figure 1 shall make this adjustment possible. It shall not be necessary to reposition the barometric setting knob along the setting knob shaft. The setting system shall be capable of correcting for zero shifts encountered in service use without requiring that the altimeter be returned to overhual or to the manufacturer for this purpose.

3.4.4.2 Setting knob position and lock. The barometric setting shall be provided with a lock which shall hold the setting knob in its normal position and allow adjustments as specified in 3.4.4. The means for unlocking the barometric setting system shall be accessible from the face of the altimeter when installed on the instrument panel. The unlocked barometric setting system shall allow adjustments as specified in 3.4.4.1. The design shall provide for a seal to prevent unauthorized adjustment. The locking mechanism shall be designed to prevent the loss of the locking screw, locking

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pin or any other part of the locking mechanism while rezeroing the altimeter. The design of the locking mechanism shall be such that in the event the locking screw is accidentally loosened, movement of the barometric setting knob to the barometric scale adjustment positions shall not be possible without further action such as pulling up on the locking screw to disengage the lock. It is intended that accidental adjustment of the relative settings of the pointer, and barometric scale counter, by the aircraft pilot is precluded in the event the locking screw becomes loose.

3.4.5 Pointer, counter and drum relationship. The pointer shall indicate readings in feet with reference to a single circular uniform graduated scale making one revolution for each 1,000 ft. of altitude. The pointer shall be positively geared to the 100 ft. drum so that as the pointer moves CW (increasing altitude), the drum will rotate for increasing numbers synchronized with the pointer. The numerals on the 100 ft. drum shall represent hundreds of feet. The 1,000 ft. drum shall be driven by the 100 ft. drum so that the 1,000 ft. drum will index one digit as the 100 ft. drum moves through the 9 to 0 position in either direction. During the long period between 0 and 9, the 1,000 ft. drum shall be locked in position and in alignment with 10,000 ft. drum. Each digit on the 1,000 ft. drum shall represent 1000 ft. intervals. The 10,000 ft. drum shall be driven by the 1,000 ft. drum so that the 10,000 ft. drum will index one digit as the 1,000 ft. drum moves through the short period between 9 and 0 in either direction. During the long period between 0 and 9 of the 1,000 ft. drum, the 10,000 ft. drum shall be locked in position. Each digit of the 10,000 ft. drum shall represent 10,000 ft. levels. In the space corresponding to zero, the 10,000 ft. drum shall have striped lines as shown in Figure 2. The 1,000 ft. drum and the 10,000 ft. drum are collectively called the counter.

The combined readings of the counter-drum shall indicate the pressure altitude in thousands and hundreds of feet. The pointer shall act as a vernier of the hundreds drum as well as being an indication of trend information. The counter-weight of the pointer shall not in any position obscure the indication of the drum.

3.4.5.1 Pointer adjustment. The pointer and drum shall be free to swing to either side of their zero position without engaging a mechanical stop. The pointer shall be firmly attached to the pointer shaft to prevent slippage under normal service use and still be removable for depot service maintenance. The pointer shall not distort and bind as a result of application of temperature differentials such as could result from sunlight on the dial or other service conditions.

3.4.6 Case. The outline dimensions and the external connections of the altimeter case shall conform to Figure 1. The body and mounting flange of the case shall be made of lightweight non-magnetic metal, uniform in texture. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish, Color No. 37038 of FED-STD-595.

3.4.6.1 Static connection. The static connection shall be in accordance with Figure 1 and shall be located as shown thereon. A 150-200 wire mesh filter screen (either stainless steel or monel) shall be firmly installed in the static connection to prevent dirt particles from entering the case. For

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storage, shipping and handling purposes, a vented plug of acceptable material and design shall be provided. The plug shall incorporate a vent consisting of a hole at least 1/16 inch in diameter. The plug shall also incorporate suitable filtering material to prevent the entry of particles which the manufacturer considers would be detrimental to the performance of the instrument.

3.5 Primary electrical input power requirements. The equipment shall meet all applicable requirements of MIL-STD-704 and give specified performance from the following power sources with characteristics as defined in MIL-STD-704, having limits as defined therein. The power required shall not exceed the specified amounts.

(1) Lighting Power 5.0 ± 0.1 volt, AC or DC, 4.0 watts maximum.

(2) 28 VDC ± 1 (Integral Vibrator) 90 ma.

3.5.1 Electrical connector. The electrical connector used shall conform to Specification MIL-C-26482 and the type specified in Figure 1 and shall be connected as specified in Figure 3. The electrical connector shall be protected with a suitable cap for shipping, storage and handling purposes.

3.5.2 Overload protection. All parts and circuits of the altimeter which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage. The use of fuses and other protective devices shall not be permitted.

3.5.3 Undervoltage protection. The altimeter shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage is within the specified limits.

3.5.4 Warmup time. The maximum warmup time for the equipment shall not exceed one minute under standard conditions and two minutes at extreme service conditions.

3.5.5 Internal vibrator. An internal vibrator shall be incorporated into the altimeter which shall effectively decrease the friction in the mechanism. The vibrator shall be self-starting upon being energized and shall start without being tapped or other external vibration assistance and shall operate under all environmental conditions specified herein. The vibrator unit shall be capable of starting, running and producing vibration sufficient to meet the friction and reliability requirements specified herein for a period of 1,000 hours without failure. The design shall be such that extended use of the vibrator will not result in deposits of loose material nor produce deleterious fumes that may result in adverse affects on altimeter performance.

3.5.5.1 Power. The internal vibrator circuit shall be designed to operate in accordance with the power requirements of MIL-STD-704, Category B (16 to 30 VDC). The input current shall not exceed 90 milliamperes. The vibrator circuit shall be capable of withstanding a reverse polarity of 28 VDC voltage without damage to the vibrator.

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3.6 Dial. The dial shall be marked as shown in Figure 2. Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595, and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. The dimensions of the dial markings shall be as shown on Figure 2. The dial shall be securely fastened to the mechanism frame by means of screws or mechanical locking and shall be of sufficient thickness that warping does not result when the instrument is subjected to the environmental conditions specified herein. The dial shall be removable at overhaul without damage.

3.6.1 Visibility of dial. The pointer, numerals, at least 1/32 inch of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 20 degrees with the perpendicular to the dial and whose small diameter is the aperture of the case.

3.6.2 Markings. All markings shall be durable so as to withstand the usage encountered in service. The style and proportions of numerals and letter placed on the dial shall conform to MS 33558.

3.6.3 Numerals. Each numeral shall distinctly indicate the graduation to which it applies. If practicable, each numeral shall be so placed that the center mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial. When several numerals are used in one group, the space between the numerals shall be approximately 1/64 inch.

3.6.4 Finish and dimensions. The following markings shall be furnished in lusterless white Color No. 37875 of FED-STD-595. The dimensions shall be as specified.

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Markings	Height or Length Inch ± 0.015	Width of Line or Graduation Inch ± 0.005
Numerals 0, 1, 2, 3, 4, 5 6, 7, 8, 9 on counters and 0, 1, 2, 3, 4, 5, 6, 9 on dial	0.281	0.035
Numerals 7, 8 on dial	0.187	0.023
100-foot graduation	0.172	0.030
50-foot graduation	0.109	0.014
Baro Counter Numerals	0.156	0.025
ALT letting	0.188	0.025
IN HG lettering	0.046	0.008
100 FT lettering	0.046	0.008
1,000 FT lettering	0.046	0.008
Shaded portion of pointer	---	---
Border on altitude counter window	---	0.035

The following markings shall be finished in lusterless black Color No. 37038 of FED-STD-595. The dimensions shall be as specified.

Markings	Height or Length Inch ± 0.015	Width of Line or Graduation Inch ± 0.005
50,000 FEET lettering	0.062	0.011
Unshaded portion of pointer, dial, counters	---	---
Numerals 0 thru 9 on 100 ft. drum	0.156	0.025
AAU-31/A-1 or AAU-31/A-2	0.062	0.011

The background of the 100 ft. drum shall be finished in lusterless gray Color No. 36622 of FED-STD-595.

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3.7 Integral lighting. The altimeter shall be provided with integral lighting in accordance with MIL-L-25467 and MIL-L-27160 to illuminate the display. The lighting colors shall be as specified in MIL-C-25050 and in 1.2.

a. Brightness for the "aviation red" and instrument and panel white lighting shall be as specified in MIL-L-25467 except as modified in (c). Applied voltage shall be as specified in MIL-L-25467.

b. Brightness for the blue-white lighting shall be as specified in MIL-L-27160 except as modified in (c). Applied voltage shall be as specified in MIL-L-27160.

c. Brightness for the grey drum (100 ft. drum) shall be 0.8 ± 0.4 foot lamberts.

The lighting circuit shall operate from 5 V AC or DC and shall be connected as in Figure 3. The integral lighting wiring shall not be grounded to the instrument case. The bulbs shall be readily replaceable, without soldering, by removing the bezel.

3.7.1 Coverglass and lighting wedge. The coverglass and/or lighting wedges shall conform to MIL-G-81704 Type I and shall not interfere with the readability of the instruument. It shall be properly sealed and shall be replaceable by removing the bezel. Provisions shall be made for alignment of the lighting wedges in the bezel to facilitate assembly at overhaul. Provisions shall also be made for procurement and replacement of individual wedges in a wedge assembly. Wedge assemblies shall be interchangeable. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-14806.

3.7.2 Distance from dial to coverglass. The distance from the dial to the innermost glass surface shall not exceed 0.250 inch.

3.8 Radio noise interference. The altimeter shall be provided with such necessary noise suppression features that it will meet the limits specified for conducted and radiated radio noise in MIL-STD-461 Class A1B, Requirements CE01, CE03, CE07, CS01, CS02, CS06, RE01, RE02, RS01, RS02, RS03. In particular, these provisions shall include adequate filters and enclosed case construction which will minimize leakage through joints, seams and mating surfaces. The volume and weight of the filtering components required shall be minimized by the application of proper electrical and mechanical design and construction.

3.9 Service conditions. The instrument shall meet the performance specified herein under the following conditions:

a. Temperature - Continuous operation at temperatures ranging from -54° to +71°C and storage at temperatures ranging from -62° to +71°C.

b. Humidity - Relative humidity up to 100 percent.

c. Altitude - Pressure altitudes ranging from -1,000 to 50,000 feet.

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d. Salt Spray - Exposure to a moist salt laden atmosphere.

e. Vibration - Vibration in accordance with 4.7.7.3.

f. Fungus - Fungus growth as encountered in tropical climates.

g. Sand and Dust - Airborne sand and dust particles as encountered in desert operations.

h. Shock - Shock in accordance with 4.7.7.5.

i. Pressurization - Compartment pressurization up to an absolute pressure of 60 in. Hg.

j. Acceleration - Acceleration forces of 20g for a period of 30 seconds.

k. Rates of climb and descent - Up to 30,000 FPM with accurate altitude provided. No damage to the altimeter shall occur at rates to 50,000 ft. per minute.

3.10 Reliability program. The contractor shall establish a reliability assurance test program in accordance with MIL-STD-781.

3.10.1 Operational stability. The altimeter shall operate with satisfactory performance, continuously or intermittently, for a period of at least 1,000 hours without the necessity for internal adjustment.

3.10.2 Reliability in Mean Time Between-Failure (MTBF). The altimeter shall have a minimum acceptable MTBF 0-1 of 1,000 hours as defined in MIL-STD-781. (0-0 = 2000 hours.)

3.10.3 Longevity. The altimeter shall have a minimum operating life of 3,000 hours with reasonable servicing and replacement of parts. Any parts requiring scheduled replacement within this time interval shall be specified by the contractor.

3.11 Identification markings. Identification markings shall be in accordance with MIL-STD-130. A nameplate shall be securely attached to the exterior of the case as shown in Figure 1. A decal bearing altimeter warranty information shall also be securely attached to the altimeter case as shown in Figure 1 when the procurement document specifies that a warranty is applicable.

3.12 Weight. The weight of the complete assembled altimeter, without connecting fittings and mounting screws, shall not exceed 2.5 pounds.

3.13 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may effect the interchangeability, operation or performance of the altimeter, they shall be held or limited accordingly.

3.14 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall

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be governed by the drawing number requirements of DOD-D-1000/1. Such changes must be reported to the qualifying agency as required for retention of qualification (see 3.1.1) and as specified for Sampling Plan B Production Samples by 4.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and test. Unless otherwise specified herein or in the contract or purchase order, the supplier is responsible for the performance of all tests. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of test. Tests of the altimeter shall be classified as follows:

a. Qualification Tests. Qualification Tests consist of examinations and tests performed on sample indicators as required for approval as a qualified product.

b. Quality Conformance Tests. Quality Conformance Tests are the examinations and tests which are applicable to indicators manufactured and submitted for acceptance under contract.

4.2.1 Qualification tests. Application for qualification shall be made in accordance with the Department of Defense Standardization Program Publication SD-6, Provisions Governing Qualification. The information required to be supplied to the Government as specified by Publication SD-6, Section 106, shall include engineering data, a complete parts list, and such detailed plans and assembly drawings necessary to identify the altimeter to be qualified and as necessary to perform the examination of product test as specified herein by paragraph 4.5.1. Unless otherwise specified, the supplier is responsible for the performance of all qualification tests. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Prior to initiating qualification testing, the manufacturer shall submit a detailed test procedure to the Government for approval. The test procedure shall be prepared in accordance with MIL-T-18303. The qualification tests consist of examinations and tests performed on sample indicators submitted for approval as a qualified product. These tests are specified under 4.5, Individual Tests, 4.6, Sampling Plan A Tests, 4.7, Sampling Plan B Tests, 4.8, Maintainability Demonstration and 4.4.3.1, Reliability Qualification Phase Tests. When the qualification tests are performed by the manufacturer, the specimens subjected to the sampling plan B tests are to be submitted to the Government laboratory designated in the letter of authorization. These samples will be subjected to limited verification tests, examination, and/or compatibility tests as deemed necessary.

4.2.1.1 Qualification test sample identification. The qualification test samples shall be plainly identified by durable tags, securely attached and marked with the following information:

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Qualification Test Sample

Altimeter Type AAU-31/A-1 (or type AAU-31/A-2)
Submitted by (manufacturer's name, date)
For qualification test in accordance
with Specification MIL-A-81851B under
authorization (reference letter authorizing tests).

4.2.2 Quality conformance test. The contractor shall furnish all samples and shall be responsible for accomplishing all tests of indicators manufactured and submitted for acceptance under contract, except Sampling Plan B tests which will be performed at a government laboratory designated by the procuring activity. The contractor may witness any tests performed by the government laboratory. Prior approval should be obtained for such visits. The contractor should also advise the government testing activity at least two weeks prior to the submission of test samples in order to expedite scheduling of tests. Quality conformance tests, except for Sampling Plan B, shall be under the supervision of the Government Quality Control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification (including detailed results of the examination of product test with respect to meeting each applicable specification requirement), signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following tests as specified herein:

a. Individual Tests (see 4.3)

b. Sampling Plan A Tests (see 4.4.1.1)

c. Sampling Plan B Tests (see 4.4.2.2)

d. Reliability Assurance Tests (see 4.4.3)

e. Longevity Tests (see 4.4.3.3)

4.3 Individual tests. Each altimeter produced under contract or work order submitted for acceptance shall be subjected to the individual tests to determine compliance with the requirements of material, workmanship and operational adequacy. As a minimum each instrument accepted shall have passed the following tests:

a. Examination of Product (4.5.1)

b. Run-in (4.5.2)

c. Barometric Setting Scale* (4.5.3)

d. Barometric Setting Knob Functional Test* (4.5.3.1)

e. Case Leakage* (4.5.4)

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f. Friction Stop and Jump* (4.5.5)

g. Static Friction* (4.5.5.1)

h. Position Error* (4.5.6)

i. Altitude Display Scale Error* (4.5.7)

j. Hysteresis and After Effect* (4.5.7.1)

k. Scale Error at -35°C* (4.5.8)

l. Scale Error at +45°C* (4.5.9)

m. Lighting (Individual)* (4.5.10)

NOTE: * Tests may be performed in any order

4.4 Sampling plans. The production sampling plans shall provide for Sampling Plan A, Sampling Plan B, and Reliability Sampling Phase Test of MIL-STD-781 as specified in 4.4.3.2. The test samples which have been subjected to only Sampling Plan A tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all of the individual tests. Test samples which have been subjected to the Sampling Plan B test shall not be delivered on contract as usable production equipment, but shall be returned, as is, to the contractor.

4.4.1 Sampling Plan A sample selection. Sampling Plan A samples shall be selected from production at random in accordance with the following schedule:

Quantity Offered for Acceptance	Quantity to be Selected for Tests
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

If the lot size is less than 25 instruments, the quantity produced in two months shall be considered as one lot. Unless otherwise noted by contract concurrent delivery shall be permitted during the storage period. When an instrument failure occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of each 15 from those still on hand and proceed with sampling as indicated above when the next quantity is offered for acceptance. In the event that a failure is noted for the sample which is subjected to the shelf life test only, the shelf life test is the sole test to which the increased sample must be subjected.

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4.4.1.1 Sampling Plan A tests. Each sample selected for Sampling Plan A test shall be subjected to the following tests:

a. Dielectric Strength (4.6.1)

b. Individual Tests (4.6.2)

c. Scale Error at -54°C* (4.6.3)

d. Scale Error at +71°C* (4.6.4)

e. Vibration Error* (4.6.5)

f. Power Variation* (4.6.6)

g. Lighting Tests (Sampling Plan A Lighting) (4.6.7)

h. Response Test* (4.6.8)

i. Baroset Knob, Torque, Push and Pull and Bending Test* (4.6.9)

NOTE: * Tests may be performed in any order

4.4.1.2 Shelf life test. One additional sample, selected at random, shall be tested from each monthly lot for shelf life test. If the lot size is less than 25 instruments, the quantity produced in two months shall be considered one lot. The altimeter shall be placed in storage for a period of not less than 30 days. During the 30 day period, the instrument shall remain as dormant as practical. The mechanism shall not be exercised other than that caused by normal atmospheric pressure variation. The temperature shall remain constant; however, normal room temperature variations will be permitted. At the end of the 30 day period, the altimeter shall be subjected to and meet the requirements of the following test in the order shown.

a. Barometric Setting Scale (4.5.3)

b. Case Leakage (4.5.4)

c. Friction Stop and Jump (4.5.5)

d. Static Friction (4.5.5.1)

e. Position Error (4.5.6)

f. Altitude Display Scale Error (4.5.7)

g. Hysteresis and After Effect (4.5.7.1)

h. Lighting (individual) (4.5.10)

This test shall be omitted for the last month's production of any contract with extended delivery of a four month period. Rezeroing of the barometric setting mechanism shall not be required to meet scale error tolerances.

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4.4.2 Sampling Plan B instructions. Three production samples shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. Unless otherwise directed, these samples shall be forwarded at the contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information.

Altimeter AAU-31/A-1 (or AAU-31/A-2)
Submitted by (manufacturer's name, date) for
production acceptance Sampling Plan B tests of
MIL-A-81851 with Contract No._____
Manufacturer's part number_____

Along with the test sample, the manufacturer shall provide a letter of certification which states therein that the test samples do not differ from the currently qualified design in any detail such as materials, parts, or assembly. When any details or design changes have been incorporated or have altered the qualified design in any way, the manufacturer shall submit a complete disclosure of the changes made. When such changes have affected the drawings, parts lists, or other documentation submitted to the government for qualification, the manufacturer shall submit updated documentation as necessary.

4.4.2.1 Sampling Plan B approval. Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished. If the manufacturer has furnished acceptable production altimeters in accordance with the specification within a two year period prior to the date of a new contract, then the Government may elect to abbreviate or omit any of the Sampling Plan B tests listed. The selection of the tests to be performed will be decided on the basis of the qualification test data, the test results of the Sampling Plan B tests performed under the prior contract, service experience, and the letter of certification submitted with the samples. Approval for delivery of production altimeters may be granted after completion of the limited tests. The Government may elect to perform the remaining unperformed tests after production approval to obtain information on possible minor deficiencies.

4.4.2.2 Sampling Plan B tests. Each sample selected for Sampling Plan B test shall be subjected to the following tests:

a. Sampling Plan A Tests (4.4.1.1)

b. Lighting (Sampling Plan B)* (4.7.8)

c. Magnetic Property* (4.7.2)

d. Radio noise interference* (4.7.3)

e. Overpressure Test* (4.7.4)

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f. Underpressure Test* (4.7.5)

g. Pressurization Test* (4.7.6)

h. Low Temperature Exposure* (4.7.7.1)

i. High Temperature Exposure* (4.7.7.2)

j. Vibration (4.7.7.3)

k. Acceleration (4.7.7.4)

l. Impact Shock (4.7.7.5)

m. Humidity (4.7.7.6)

n. Fungus (4.7.7.7)

o. Salt Spray (4.7.7.8)

p. Sand and Dust (4.7.7.9)

q. Internal Examination (4.7.9)

NOTE: The instrument selected for examination of the internal mechanism shall not be exposed to the Fungus Test.

The tests listed shall be performed in the order shown except the asterisked tests which may be performed in any order. The Government test activity shall report to the manufacturer any failure of the test specimens in meeting specification test requirements prior to continuation of tests.

4.4.3 Reliability assurance tests. Reliability Assurance Tests shall be conducted using MIL-STD-781 as modified herein. Altimeters selected for reliability assurance tests shall first be subjected to the individual tests (see 4.3) and the lighting test of 4.6.7.

4.4.3.1 Reliability qualification phase. Prior to the acceptance of the altimeter three (3) to six (6) instruments shall be tested as outlined in MIL-STD-781 and as specified herein. Criteria for Test Plan II C shall be used. The altimeter shall demonstrate a minimum MTBF (0-1) of 1000 hours as defined in MIL-STD-781. (0-0 = 2000 hours)

4.4.3.1.1 Duty cycle. The duty cycle test profile shall be as specified below and as shown in Figure 4.

a. Exercise. The altimeter shall be pneumatically cycled from approximately sea level pressure to approximately 45,000 feet altitude and return to sea level within approximately 15 minutes for a total of at least four cycles per hour. During the external vibration period (10 minutes of each hour (see 4.4.3.1.1 e)) the cycling rate shall be increased to 20,000 feet per minute. The barometric setting knob shall be rotated at least once throughout the complete range each 24 hour period.

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b. Temperature. Temperature cycling shall be performed from -54°C to +55°C. Temperature cycling shall be the time to stabilize at -54° for two hours followed by the time to stabilize at the high temperature plus four hours as shown in Figure 4.

c. Power. During each hour, all electrical power, including power to the integral vibrator, shall be on for 57 minutes and off for three minutes. A two minute warm-up time is permitted prior to initiation of cycling.

d. Lighting. 57 minutes on, 3 minutes off each hour (red and white lighting to be on separately and alternately).

e. Vibration. During each hour, 10 minutes external vibration of $\pm$ 0.5 g at any non-resonant frequency between 20 and 60 Hz.

f. Input voltage. Nominal specified voltage +5% - 2%.

g. Test length. Testing will continue until the total unit hours together with the total count of relevant equipment failure permit either an accept or reject decision.

4.4.3.2 Reliability sampling phase tests. Samples of the altimeter shall be tested as outlined in 4.4.3.1.1. An alternative Production Reliability test may be specified by the procuring activity. When the alternative Production Reliability Test Plan is specified in the contract or purchase order, the contractor shall prepare such a plan and submit it to the procuring activity for approval. The plan shall consider the specific altimeter design and be based upon the reliability test plans of more recent production contracts. Guidelines will be specified by the procuring activity.

4.4.3.2.1 Lot size for sampling phase. Three (3) to six (6) instruments, selected at random, shall be tested from the monthly lot for the accept-reject criteria of Test Plan IV C. If the lot size is less than 25 instruments, the quantity produced in two months shall be considered as one lot. At the conclusion of each month, the test results shall be sent to the procuring activity and the Naval Air Systems Command, Attn: Avionics Division. At any time that the test results show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc.

4.4.3.3 Longevity tests. Unless otherwise specified by the procuring activity, a longevity test shall be performed in accordance with Test Plan XXVIII of MIL-STD-781 and the duty cycle herein, except that 3,000 hours operating time shall be accumulated on each of two units. Two of the altimeters used during the the reliability assurance tests shall be used for this test and all operating time accrued during reliability testing may be counted toward the longevity time. Altimeter performance checks may be performed at 500 hour intervals in lieu of 250 hour intervals after completion of 1000 hours of testing for each instrument.

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4.4.3.3.1 Performance characteristics to be measured during Longevity and Reliability Assurance Tests. The altimeter shall be observed each working day for any obvious failures such as rough pointer movement, vibrator operation or integral light failure. At the end of each 250 operating hours, the instruments shall be subjected to the friction and scale error tests of 4.5.5, 4.5.5.1 and 4.5.7 at the asterisked test points of Tables I and II. The integral vibrator shall be checked to ascertain proper starting characteristics during the static friction test of 4.5.5.1. At the completion of the reliability and longevity tests, the altimeters shall be subjected to all individual tests (except run-in, scale error at -35°C, scale error at +45°C) and the lighting test of 4.6.7.

4.4.3.4 Failure criteria. If recalibration, adjustment or repair is required to pass any of the tests listed in 4.4.3.3.1, either at the 250 hour test intervals or at the completion of the reliability test, a failure shall be charged to that test. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one part caused the other parts to fail. A 25 ft. out-of-tolerance condition in scale error will be permitted after each 250 hour cycling period, provided rezeroing by use of the barometric setting system will correct the condition. Any other out-of-tolerance condition shall be charged as a failure. Any malfunction of the integral vibrator shall be charged as a failure. (NOTE: The 25 ft. out-of-tolerance allowed for scale error after each 250 hour cycling period shall be applied as a data correction and shall not be a physical correction by means of the zero setting system.)

4.4.3.5 Preventive maintenance. No preventive maintenance shall be performed on the altimeter during the 1,000 hour MTBF test. The failures occurring during the longevity test shall be analyzed in accordance with MIL-STD-781. In addition, the analysis of failures, especially for what appears to be pattern type failures, should indicate whether preventive maintenance would have prevented the failure and extended the service life.

4.4.3.6 Disposition of samples. Any reliability sample tested may be delivered on contract provided the sample is in "good as new" condition or has been refurbished and meets the individual inspection tests and is representative of production units currently being accepted.

4.4.4 Test conditions. Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature: Room ambient, 25°C ± 5°C
Pressure: Normal atmosphere (approximately 29.92 in. Hg.)
Humidity: Room ambient 40 to 90% relative humidity
Power: 5 ± 0.1 VAC or DC (for lighting)
28 VDC ± 1 volt (for integral vibrator)

4.4.4.1 Test readings. Unless otherwise specified, vibrator and lighting power shall be applied to the altimeter during all tests. The altimeter shall not require external vibration or tapping to meet the performance specified herein. Application of external vibration in the order of 0.2 g or tapping the instrument shall not cause the instrument to exceed the applicable tolerances. The altimeter shall not require internal heaters to meet the performance requirements specified herein.

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4.4.4.2 Attitude. Unless otherwise specified, the altimeter shall be tested in its normal operating position.

4.4.4.3 Integral lighting. The altimeter lighting system shall be energized and shall operate satisfactorily during all tests when an operational requirement for the altimeter is specified. The lighting shall be tested and operate satisfactorily after completion of each non-operational test.

4.4.4.4 Rate of pressure change. Unless otherwise specified, the rate of change in pressure during all tests for both increasing and decreasing altitudes shall be accomplished at any rate up to 30,000 feet per minute. The rate shall be progressively reduced as the checkpoints are approached to avoid passing the checkpoint. The movement of the pointer during increasing and decreasing altitudes shall be smooth and free from irregular motion when the integral vibrator is operating.

4.4.4.5 Altimeter zero setting. Resetting of the altimeter zero shall not be accomplished at any time during the course of testing and shall not be required to meet the accuracy tolerances specified herein except as specified during the reliability test. It is intended that all samples submitted for acceptance also meet this requirement. Any exceptions requested by the contractor to reduce production instrument test or calibration time by making a zero adjustment after completion of acceptance tests shall establish that bias or calibration shifts did not occur during the acceptance tests and that the intended requirements are met. Adjustments may be permitted if the (average) difference in room temperature scale errors at the start of individual tests (for low altitude test points) do not differ from the scale errors recorded after completion of all individual tests by more than ten feet. The rezeroing procedures for such cases must be approved by the qualifying activity.

4.4.4.6 Standard pressure. The standard pressures used in calibrating and testing the altimeter-encoders shall be as specified in MIL-STD-859.

4.4.4.7 Altitude display. The amount of allowable jitter in the instrument pointer, caused by the integral vibrator, shall be no greater than a total of 10 feet (indicated) of pointer excursion during a time when the pressure applied to the instrument is not being changed. This allowable jitter shall not be added to other scale error tolerances.

4.5 Individual tests.

4.5.1 Examination of product.

a. Qualification tests. Prior to final assembly of the altimeter to be subjected to qualification tests, a detailed examination shall be made to verify compliance with each of the design requirements of Section 3 that are not covered by specific tests.

b. Quality conformance tests. Each altimeter submitted for acceptance shall be examined externally to determine conformance with the applicable drawings and with each of the requirements of this specification capable of being inspected in this manner.

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4.5.2 Run-in. Each altimeter shall be subjected to a run-in test of six hours duration. The first two hours of the run-in shall be accomplished with the altimeter exposed to a temperature of +71°C. The remaining four hours of the test shall be accomplished under room ambient conditions with the altimeter mounted on a special test table. The table shall subject the instrument to roll, pitch, and yaw motion (Scorsby) at a frequency of five to seven oscillations per minute through an angle of 30 degrees.

Prior to initiation of the run-in, the scale error at zero altitude (29.92 in. Hg.) shall be determined and recorded. The scale error shall be within the room temperature scale error tolerance of Table II. During the performance of the entire run-in tests, the altimeter shall be subjected to vibration of approximately 0.5 g (as measured on the instrument panel) at a frequency between 20 and 60 Hz. (External vibration need not be applied if the specified 0.5 g vibration level is achieved by use of the integral vibrator only.) The 5V lighting circuits shall be energized at all times during the test, alternating the circuit from red lighting to white lighting each hour. The power to operate the integral vibrator (28 VDC) shall also be applied continuously during the test performed at high temperature and at ambient room temperature. The altimeter shall be pneumatically cycled during the six hour run-in test. The pneumatic cycle shall range from approximately sea level pressure to about 35,000 feet altitude and return to sea level within approximately 15 minutes for a total of at least 4 cycles per hour. The barometric setting shall not have changed by more than .01 in. Hg. from the initial setting after completion of the run-in test.

Failures noted for production altimeters during the run-in test shall be summarized and reported in the monthly report submitted for production reliability tests (see 4.4.3.2.1).

4.5.3 Barometric setting scale. The altimeter shall be subjected to atmospheric pressure. The pressure scale shall be set successively to the values specified in Table III for both the clockwise and counter-clockwise directions of rotation of the baroset knob.

After the series of readings have been recorded, the difference between the reading at the setting of 29.92 and each other setting shall be found. (NOTE 1: The reading obtained at the 29.92 setting for the clockwise direction of rotation of the baroset knob shall be used to compute the differences for both the clockwise and counter-clockwise directions of rotations.) The differences shall be recorded and compared with the values in the "Correct Difference" column of Table III. The recorded differences shall not differ from the correct differences by more than $\pm$ 25 ft. at any setting. The difference between the values recorded for the clockwise and counter-clockwise rotations of the baroset knob shall not exceed 20 ft. Atmospheric pressure shall be recorded before and after this test and proper allowance shall be made for any variation. (NOTE 2: The design of the altimeter shall be such that the test requirement shall be met under any atmospheric pressure within the range -1,000 to +6,000 ft.) (NOTE 3: This test may be performed with the static port connected to a controlled pressure source if provisions are made to insure that some altimeters are subjected to controlled pressures at settings other than 29.92.)

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4.5.3.1 Barosetting knob functional test. The barometric setting system shall be operated slowly from 28.10 in. Hg. to 31.00 in. Hg. and return to 28.10 in. Hg. The torque required for operation from stop to stop throughout this range shall be measured and shall be 3-10 inch ounces.

4.5.4 Case leakage. A vacuum sufficient to produce a reading of 3,000 feet (approximately) above ambient altitude shall be applied to the static pressure port of the altimeter, at which point the connecting tubing shall be pinched off or otherwise completely sealed. The change in reading in one minute shall not exceed 100 ft. The leakage shall be checked in this manner for each 90° position of the baroset knob. The leakage check for each 90° position of the baroset knob is applicable only to room temperature tests. This test shall be repeated for an altimeter reading of 30,000 feet altitude (approximately). Altimeters shall be tested individually. This test may be combined with the Altitude Display Scale Error Test (see 4.5.7).

4.5.5 Friction stop and jump. The altimeter shall be tested for friction in both ascending and descending directions at the test points specified in Table I. With the internal vibrator energized, the pressure to the static port of the altimeter shall be applied at the rates indicated for the test points in Table I. The change in the indicating position of the pointer, due to apparent stopping and jumping as the counter is turning or at any other place around the dial, shall be recorded as friction and shall not exceed the tolerances specified in Table I, Column C. This test shall be repeated with all electrical power off. The stop and jump friction for the test as specified shall not exceed the tolerances specified in Table I, Column D. After the preceding requirements are met, the altimeter shall again be subjected to the vibrator operative test, except the rate-of-change of altitude shall be 20,000 FPM and the face of the unit shall be inclined down 45° during ascent and inclined up 45° during descent. The tolerance during this portion of the test shall be increased by 50 ft. over the values listed in Column C of Table I. The barometric pressure setting may be set at any point within its range during this test.

NOTE: When the data from a production lot indicates that the portion of the test performed for 45° inclination angles and at 20,000 feet per minute altitude change rates is satisfactory for every instrument of the lot, the contractor may perform this test on a sampling plan basis only. In the event of failure to meet the requirements for this portion of the test during Sampling Plan A or Sampling Plan B tests, the contractor shall immediately revert to 100% tests of all production units.

4.5.5.1 Static friction. The altimeter shall be subjected to this test with all electrical power off. A decreasing pressure not exceeding 3000 fpm shall be applied to the static pressure port. As the static friction test point listed in Table I is approached, the rate of pressure change shall be gradually reduced until the desired test point is attained. The pressure shall then be held constant while two readings of the altimeter are taken, the first without the integral vibrator operating and the second with the vibrator operating. The difference between the two readings shall be recorded as static friction and shall not exceed the tolerances listed Table I.

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4.5.6 Position error. The altimeter position error readings shall be taken while the instrument is vented to atmosphere and in each of the following positions:

a. Normal operating position

b. Rotated about the longitudinal axis 90°

c. Rotated about the longitudinal axis 180°

d. Rotated about the longitudinal axis 270°

e. Dial face up.

The readings in any test position shall not differ from the reading recorded in any other test position listed by more than 20 feet. This test shall be performed with 28 VDC vibrator power applied to the altimeter. The test procedure shall permit tests of altimeter-encoders with the pointer positioned at any point around the dial.

4.5.7 Altitude display scale error. For a period of not less than 6 hours prior to this test, the altimeter shall not have been subjected to any testing involving operation at other than atmospheric pressure. The altimeter shall be tested for scale errors as follows: The barometric setting shall be set at 29.92 in. Hg. and shall remain at this setting during performance of all scale error tests. The direction at which the 29.92 in. Hg. setting point is approached shall not be a factor in determining scale error accurancy. The resultant scale errors shall be within the specified tolerances regardless of the direction of approaching the setting.

The instrument shall then be subjected throughout the applicable range to successively decreasing pressure and then increasing pressure at the test points specified in Table II. The instrument shall remain at the pressure corresponding to each test point for at least one but not more than five minutes before a test reading is taken. The scale errors shall not exceed the tolerances specified in Table II. This test may be performed with the instrument in an altitude chamber. The movement of the altimeter-encoder counter-drum-pointer display shall be free from apparent backlash and any irregular motion when the pressure is varied uniformly.

4.5.7.1 Hysteresis and after effect. Tests of this paragraph may be combined with the Altitude Display Scale Error test (see 4.5.7). Not more than 15 minutes after the altimeter has been first subjected to the pressure corresponding to 50,000 ft. in the test specified in 4.5.7, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 20,000 FPM until the pressure corresponding to the test point given in Table IV is reached. The instrument shall remain at this pressure for at least one but not more than 15 minutes before the test reading is taken. The pressure shall again be increased at the same rate until the pressure corresponding to the second test point in Table IV is reached. The instrument shall remain at this pressure for at least one but not more than ten minutes before the test reading is taken. The reading of the instrument

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at the three test points shall not differ from the reading of the instrument for the same test points in the tests specified in Table IV. In addition, all test readings shall be within the tolerance of Table II.

4.5.8 Scale error at -35°C. The altimeter, without electrical power applied, shall be subjected to a temperature of -35°C for a period of three hours. At the end of this time period, all electrical power shall be applied to the altimeter, and the Case Leakage (see 4.5.4), and Altitude Display Scale Error (see 4.5.7) tests shall be repeated while a temperature of -35°C is maintained. The test points and tolerances shall be as specified in Table II. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error at the zero altitude test point only shall be determined. The room temperature scale error tolerance shall apply for this check. During production testing, the scale error at zero altitude check need not be performed if the scale error at 45°C test per 4.5.10 is performed subsequent to the test at -35°C.

4.5.9 Scale error at +45°C. The tests specified in 4.5.8 (Scale error at -35°C) shall be repeated except that the temperature shall be +45°C. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error at the zero altitude test point only shall be determined. The room temperature scale error tolerance shall apply for this check. During production testing, the scale error at zero altitude check need not be performed if the scale error at -35°C test per 4.5.8 is performed subsequent to the test at +45°C.

4.5.10 Lighting. The lighting system shall be excited with 5.0 $\pm$ 0.1V and examined to verify that all lamps function properly. Both the red lighting system and the white lighting system shall be checked.

4.6 Sampling plan A tests

4.6.1 Dielectric strength. The altimeter shall be checked for insulation resistance by testing from the terminals indicated to ground at the potentials indicated for no more than 30 seconds.
NOTE: The maximum voltage shall be raised from zero by rheostat control; switching to the voltage from zero shall not be permitted.

MP	together	500V 60Hz
XYZ	together	500V 60Hz

There shall be no breakdown of insulation or any other permanent damage to the indicator as a result of this test.

4.6.2 Individual tests. The tests specified in 4.3 shall be performed.

4.6.3 Scale error at -54°C. The tests specified under 4.5.7 (Altitude Display Scale Error) shall be repeated except that during the test and for a period of 3 hours without power applied before the test, the altimeter shall be at a temperature of -54°C $\pm$ 2°C. The test shall be initiated within two minutes after power is applied. The test points and tolerances shall be as specified in Table II. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error at the zero altitude test

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point only shall be determined. The room temperature scale error tolerance shall apply for this check. During production testing, the scale error at zero altitude check need not be performed if the scale error at +71°C is performed subsequent to the test at -54°C.

4.6.4 Scale Error at +71°C. The test specified in 4.6.3 (Scale Error at -54°C) shall be repeated except the test temperature shall be +71°C ± 2°C. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error at the zero altitude test point only shall be determined. The room temperature scale error tolerance shall apply for this check. During production testing, the scale error at zero altitude check need not be performed if the scale error at -54°C is performed subsequent to the test at +71°C.

4.6.5 Vibration error. The altimeter shall be mounted on the vibration stand in its normal operating position and secured by flange screws in the same position as on an aircraft panel. The barometric pressure setting shall be recorded. The instrument shall be subjected to one hour vibration with an amplitude between 0.003 and 0.005 inch at a frequency which is varied uniformly from 500 CPM to 3,000 CPM and return to 500 CPM during the hour. The vibration shall be such that any point on the instrument case will describe a circular path of the amplitude stated, inclined at an angle of 45° to the horizontal and dial planes. The natural frequency of the instrument shall not occur in this frequency range. While the instrument is being vibrated, the maximum amplitude of the pointer oscillation shall not exceed 20 ft. and the pointer variation from normal shall not exceed 20 ft. No shift in the barometric setting counter indication shall occur with respect to the setting recorded prior to vibration. The integral lighting of the instrument shall be operated during this test and shall not flicker. While the instrument is being vibrated, the drum shall not exhibit oscillatory motion greater than 20 ft.

Upon completion of the preceding vibration cycling, the one hour test shall be repeated except that the amplitude of vibration shall be set between 0.009 and 0.011 inch. Under this test condition,the maximum amplitude of pointer oscillation shall not exceed 30 feet and the pointer variation from the normal shall not exceed 30 feet. No shift in the barometric scale counter shall occur with respect to the setting recorded prior to vibration. A change in the barometer setting by 1/2 digit (.005 in. Hg) during or after the performance of the vibration test is defined as a shift. The integral lighting of the instrument shall be operated during this test and shall not flicker. At the completion of the vibration error test, the scale error at zero altitude test point shall be determined. The room temperature scale error tolerance shall apply for this check. All individual tests listed in 4.6.10 shelf life, must be performed after completion of the vibration test if for any reason all tests samples will not be subjected to the shelf life test.

4.6.6 Power variation. The vibrator excitation voltage shall be reduced to 20 volts and the altimeter subjected to the friction stop and jump test. The asterisked test points of Table I shall be used and the tolerances of Column "C" shall apply. The vibrator excitation voltage shall be reduced to 16 volts for a period of one minute. There shall be no damage to the vibrator. When the excitation voltage is returned to normal, the vibrator shall operate

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satisfactorily. The polarity of the vibrator excitation voltage shall then be reversed for a period of one minute. There shall be no damage to the vibrator or associated electrical circuitry as a result of this test.

4.6.7 Lighting tests. Lighting tests specified as Sampling Plan A tests in MIL-L-25467 and MIL-L-27160 shall be conducted. Those tests which are common may be conducted once.

4.6.8 Response test. The altimeter shall be subjected to a rate of change of pressure equivalent to 30,000 ft/min. No damage to the internal mechanism shall occur due to this test and no zero or calibration shift shall result. Operation of the counter-drum-pointer display shall be free from irregular motion. The altimeter shall then be subjected to rates of change of pressure equivalent to 50,000 ft/min. without damage to the internal mechanism. At the completion of this test, the altimeter shall be checked for scale error at the zero test point.

4.6.9 Baroset knob torque, push and pull and bending test. The following tests shall be performed to demonstrate that the design meets the requirements specified in 3.4.4 (barometric Setting System).

a. The setting range and presence of the required stops shall be verified. A twisting force of 6 pound inches maximum shall be applied at the barosetting knob against each stop. The stops shall be verified to be located as required by 3.4.4 and the mechanism shall not be damaged or the calibration affected by the torque applied. When a mechanism is incorporated in the instrument to limit the torque applied to the stops, the torque limit prior to slip should be recorded and satisfactory operation of the mechanism demonstrated. The travel of the altitude pointer for one turn of the barometric setting knob shall be within the range of 200 to 300 feet. Clockwise rotation of the setting knob shall result in clockwise rotation of the pointer.

b. A pulling force of 15 pounds shall be applied to the barosetting knob. While the force is applied, the knob shall be rotated at least 90°. This action shall be repeated with a pushing force of 15 pounds applied. The relationship of the display of the barometric setting indication shall not be affected by the forces applied and the mechanism shall not be otherwise damaged.

c. The barosetting knob shall withstand a force of 15 pounds applied at the extreme outward end of the knob without permanent deflection or damaging of the mechanism.

4.7 Sampling Plan B.

4.7.1 Sampling Plan B tests. The Sampling Plan A tests specified in 4.4.1.1 shall be performed.

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4.7.2 Magnetic property. The altimeter, not energized, shall be rotated in a vertical plane about a short-bar magnetic compass with the nearest part of the altimeter 5.5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the instrument shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315° from the initial position. At each of these positions, the instrument shall be rotated about on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. With the altimeter at any specified position, the compass deflection shall not exceed 3°. This test shall be repeated with the instrument energized with rated voltage.

4.7.3 Radio noise interference. The altimeter shall be subjected to a radio noise interference test per MIL-STD-461 Class A1B, Requirements CE01, CE03, CE07, CS01, CS02, CS06, RE01, RE02, RS01, RS02, RS03. Tests shall be performed with the integral vibrator and lighting operating.

4.7.4 Overpressure test. With the barometric setting set at 29.92, a pressure of 35 in. Hg. absolute shall be applied to the static port of the altimeter for a period of one minute. No damage to the internal mechanism shall occur due to this pressure and no zero or calibration shift shall result. At the completion of this test, the altimeter shall be checked for scale error at the zero altitude test point.

4.7.5 Underpressure test. The altimeter shall be subjected to an altitude of 60,000 ft. for a period of one minute. No damage to the internal mechanism shall occur due to this pressure and no zero or calibration shift shall result. At the completion of this test, the altimeter shall be checked for scale error at the zero altitude test point.

4.7.6 Pressurization test. The altimeter shall be placed in a pressure chamber with the pressure port connected to outside ambient pressure (29.92 inches of mercury approximately) during this test. The chamber pressure shall be raised to 60 in. Hg. absolute and maintained at approximately 25°C for a period of 4 hours. During the 4 hour period, no deformation or other damage to the instrument shall occur. At the completion of this test, the altimeter shall be checked for scale error at the zero altitude test point.

4.7.7 Environmental tests. The altimeter shall be subjected to the following tests to be performed in accordance with the specified procedure of MIL-STD-810. Upon completion of each test, the altimeter shall be subjected to and meet the tests of 4.5.4 (Case Leak), 4.5.5 (Friction Stop and Jump), 4.5.5.1 (Static Friction), 4.5.7 (Altitude Display Scale Error), and 4.5.10 (Lighting). The asterisked test points of Tables I and II shall apply. When external connections are required to simulate installed conditions, they shall consist of standard AN or MS fittings and at least 10 ft. of 1/4 inch or 3/8 inch tubing, coiled and so arranged as to allow drainage of condensed vapors. The tubing shall be attached to the static pressure port of the instrument.

4.7.7.1 Low temperature exposure. The low temperature exposure test shall be performed with Method 502.2, Procedures I and II except as follows:

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a. Delete steps 6 through 9 in Procedure I.

b. Use a minimum temperature of -54°C for a period of 24 hours for Procecure I.

c. Use a minimum temperature of -54°C for the operating temperature in Procedure II.

d. The operation tests at -54°C shall consist of altitude display scale error tests.

4.7.7.2 High temperature exposure. The high temperature exposure test shall be performed in accordance with Method 501.2, Procedures I and II except as follows:

a. Delete steps 5 and 6 in Procedure I.

b. Delete steps 2, and 12 through 14 in Procedure II.

c. Use a maximum temperature of 71°C for Procedures I and II.

d. The operation tests at 71°C shall consist of altitude display scale error tests.

4.7.7.3 Vibration. The vibration test shall be performed in accordance with Method 514.3, Category 10, Description b, Test Procedure I, Test Conditions I-3.2.12, with the following modifications:

a. The 2g portion of Figure 514. 3-35 shall be extended through to 2000 Hz.

b. For the frequency range from 5-500-5 Hz., the sinusoidal cycling time shall be 30 minutes and the sweep time shall be 30 minutes.

c. For the frequency range from 500-2000-500 Hz., the sinusoidal cycling time shall be 30 minutes and the sweep time shall be 8.5 minutes.

During this test, the altimeter shall be energized and the pressure input to the altimeter shall be varied from approximately sea level to about 10,000 - 15,000 feet altitude and returned to sea level. The rate of change of pressure shall be approximately 2000 - 4000 feet per minute. During vibration, the barometric setting counter shall not change by more than 0.01 in Hg. with respect to the setting recorded prior to the test. The plane of the dial shall be vertical during all tests. If resonance is noted during the sinusoidal cycling, a resonance dwell of 10 minutes shall be performed at the frequency for which maximum resonance is observed.

4.7.7.4 Acceleration. The acceleration test shall be performed in accordance with Method 513.3, Procedure I, except that the altimeter shall be mounted in its normal operating position with the rear of the case toward the center of rotation and shall be subjected to an acceleration of 20g for a period of one

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minute. This test shall be repeated with the instrument mounted successively in the following two positions: (1) rotated 90° in azimuth with respect to the original position, (2) dial horizontal.

4.7.7.5 Impact shock. The shock test shall be performed in accordance with Method 516.3, Procedure I, figure 516.3-4, amplitude a, time duration c for operational test.

4.7.7.6 Humidity. The humidity test shall be performed in accordance with Method 507.2, Procedure III except the test item may be removed from the chamber at the completion of the exposure period for the performance of tests. The altimeter shall be thoroughly examined within at least 48 hours after completion of the humidity exposure and there shall be no evidence of corrosion and deterioration that would affect subsequent operation of the instrument.

4.7.7.7 Fungus. The fungus test shall be performed in accordance with Method 508.2, Procedure I. At the end of the test period, the altimeter shall be thoroughly examined to ascertain that no fungus growth has occurred.

4.7.7.8 Salt spray. The salt spray test shall be performed in accordance with Method 509.2, Procedure I. The altimeter shall be subjected to the tests listed in 4.7.7 (Environmental Tests) immediately after completion of the salt spray exposure period and again 48 hours (approximately) later.

4.7.7.9 Sand and dust. The sand and dust test shall be performed in accordance with Method 510.2, Procedure I.

4.7.8 Lighting. Lighting tests shall be conducted in accordance with Sampling Plan B tests of MIL-L-25467 and MIL-L-27160 as specified. Those tests which are common to both specifications may be conducted once. Integral lighting colors shall be in accordance with MIL-C-25050.

4.7.9 Internal examination. Only one instrument shall be subjected to this test. The barometric setting locking screw of the altimeter shall be loosened and it shall be demonstrated that the barometric setting system meets the requirements specified by 3.4.4.1 and 3.4.4.2. The range of adjustment of the barometric scale counter relative to the altitude pointer shall be recorded. The case of the altimeter shall then be opened and the internal mechanism shall be examined to determine conformance with the applicable requirements of the specification. The lamp circuit of the altimeter shall also be examined for conformance to the requirements of MIL-L-25467. In addition, any damage or deterioration to the mechanism that resulted from any of the preceding tests, or any other deficiency which could in any manner prevent the instrument from meeting operational or maintenance requirements during service life, shall be cause for rejection. The qualification sample altimeter shall be re-cased and demonstrated to perform satisfactorily during abbreviated room temperature tests. Such instruments may be cleaned, prior to re-casing.

4.7.10 Rejection and retests. When any altimeter selected for Sampling Plan B or Reliability Tests fails to meet the specification, no altimeters on hand or later produced shall be accepted until the extent and cause of failure are determined and appropriately corrected. Such action is also applicable in the

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event that samples submitted to the Government for Sampling Plan B tests do not meet specification. The manufacturer shall provide a report to the government representative explaining the cause of failure and the action taken to preclude recurrence. After corrections have been made, all necessary tests shall be repeated. When a failure requires the general application of revised procedures in production or the incorporation of corrective measures for other instruments in production or on hand the Government may elect to replace the original samples with new samples selected at random from the quantity offered for acceptance. New samples will be selected when the correction accomplished is of a nature that the original samples are not sufficiently representative of production.

If necessary to demonstrate satisfactory performance, all items still on hand shall be subjected to the test which produced the failure. Shipments shall not be authorized until it is determined that the instruments on hand are free of the defect detected.

4.8 Maintainability demonstration. The maintainability demonstration shall be in accordance with MIL-STD-471, Test Method 1, Plan B1. The maintainability demonstration shall be performed on a minimum of three selected repairs. The tasks selected to demonstrate maintainability shall be selected in advance and shall be approved by the procuring activity. An equivalent maintainability demonstration may be performed prior to initiation of qualification testing on approval of the qualifying activity.

5. PACKAGING

5.1 Packaging. The altimeter shall be packaged in accordance with MIL-STD-794, Level A or C, as specified in the contract or order. For Level A packaging, the method of preservation shall be in accordance with Specification MIL-P-116, Method 1A-5 without preservation compound, using reusable metal containers.

5.1.1 Shipping carton. Each altimeter shall be provided with a protective shipping carton to minimize damage during storage, shipping, and handling. The protective shipping carton shall be made of polystyrene or other suitable material to protect the entire instrument including dial glass, bezel, and protruding barometric setting knob from damage. The carton shall be marked providing for identification of the instrument as an altimeter and state that the shipping carton is reusable. The design of the carton shall be approved by the procuring activity.

5.2 Packing. The altimeter shall be packed in accordance with MIL-STD-794, Level A, B or C as specified in the contract or order.

5.3 Marking. The interior and exterior containers shall be marked as specified in MIL-STD-794 with the additional precautionary marking appearing conspicuous on two opposite sides of each interior package and each shipping container: "FRAGILE - DELICATE INSTRUMENTS - HANDLE WITH CARE."

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6. NOTES

6.1 Intended use. The altimeter covered by this specification is intended for use in helicopters and moderate performance aircraft to indicate the pressure altitude of the aircraft above some reference point, assuming standard conditions of temperature and pressure.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, revision, and date of this specification.

b. Altimeter type (see 1.2).

c. Whether Sampling Plan B is to be omitted, or the name of the laboratory where samples shall be submitted. Also, the disposition of samples (see 4.4).

d. Whether Reliability Qualification Phase tests are required (see 4.4.3 and 4.4.3.1). Whether MIL-STD-781 or alternative Reliability Test is to be used.

e. Reliability Sampling Phase tests (see 4.4.3 and 4.4.3.2 (specify MIL-STD-781 or alternative test plan),

f. Longevity Tests (see 4.4.3.3).

g. Maintainability Demonstration (see 4.8).

h. Levels of packaging and packing desired (see 5.1 and 5.2).

i. Protective shipping carton required (see 5.1.1).

j. Warranty decal information, if applicable (see 3.11).

6.3 Definitions.

6.3.1 Good as new. "Good as new" shall mean the indicator has been operated less than 10 percent of specified MTBF operation. Prior to delivery, the indicator shall be subjected to and meet all the requirements of a new indicator, except the run-in test need not be performed.

6.3.2 Refurbished. "Refurbished" shall mean the indicator has been completely overhauled with all component parts meeting current parts standards; and prior to delivery the indicator shall have been subjected to, and met, all requirements of a new indicator.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL 81851) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be

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awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Avionics Center, Indianapolis, Indiana, and information pertaining to qualification of products may be obtained from that activity.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I

STOP AND JUMP FRICTION ERROR

COLUMN "A" Test Point Altitude (Ft)	COLUMN "B" Rate of Ascent Or Descent (Ft/Min)	COLUMN "C" Tolerance Vibrator Operative (Ft)	COLUMN "D" Tolerance Vibrator Inoperative (Ft)
* 0	500	25	300
1,000	500	25	120
3,000	500	25	120
* 5,000	500	25	120
*10,000	500	25	350
20,000	3,000	25	350
*30,000	3,000	25	350
35,000	3,000	50	600
*40,000	3,000	50	600

NOTE: Tolerance changes 500 ft. below test points 00; 10,000; and 30,000 (i.e., down to -500 ft., down to 9,500 ft., and down to 29,500 ft)

STATIC FRICTION ERROR-VIBRATOR INOPERATIVE

Test Point Altitude (Ft.)	Tolerance (Ft.)
* 500	70
1,500	70
2,500	70
3,500	70
*7,500	80
12,500	90
17,500	110
22,500	120
*32,500	180
35,500	300
*42,500	300

NOTE: Asterisked points apply for the environmental and reliability tests as specified herein.

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TABLE II

SCALE ERRORS

Equivalent Pressure In. Hg. at 0°C	Standard Altitude Feet	Scale Error Tolerances (Feet) -35°C	Room Temp. and +45°C	-54°C and +71°C
31.018	-1,000	±45	±30	
*29.921	0	45	30	±60
29.385	500	50	35	
28.856	1,000	50	35	
*27.821	2,000	60	40	80
26.817	3,000	70	45	
*24.896	5,000	75	50	100
22.225	8,000	80	60	
*20.577	10,000	85	65	130
19.029	12,000	90	70	
16.886	15,000	100	80	
14.942	18,000	105	90	
*13.750	20,000	110	95	190
11.103	25,000	120	105	
8.885	30,000	135	115	
* 7.041	35,000	150	125	250
* 5.538	40,000	160	140	280
4.355	45,000	205	185	
3.425	50,000	500	500	

NOTE: Asterisked points apply for the environmental and reliability tests as specified herein.

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TABLE III

BAROMETRIC SETTING SCALE

Pressure Scale Setting	Correct Difference
28.1	-1,727
28.5	-1,340
*29.0	-863
*29.5	-392
*29.92	0
30.5	531
*30.9	893
31.0	983

NOTE: Asterisked points apply to environmental tests.

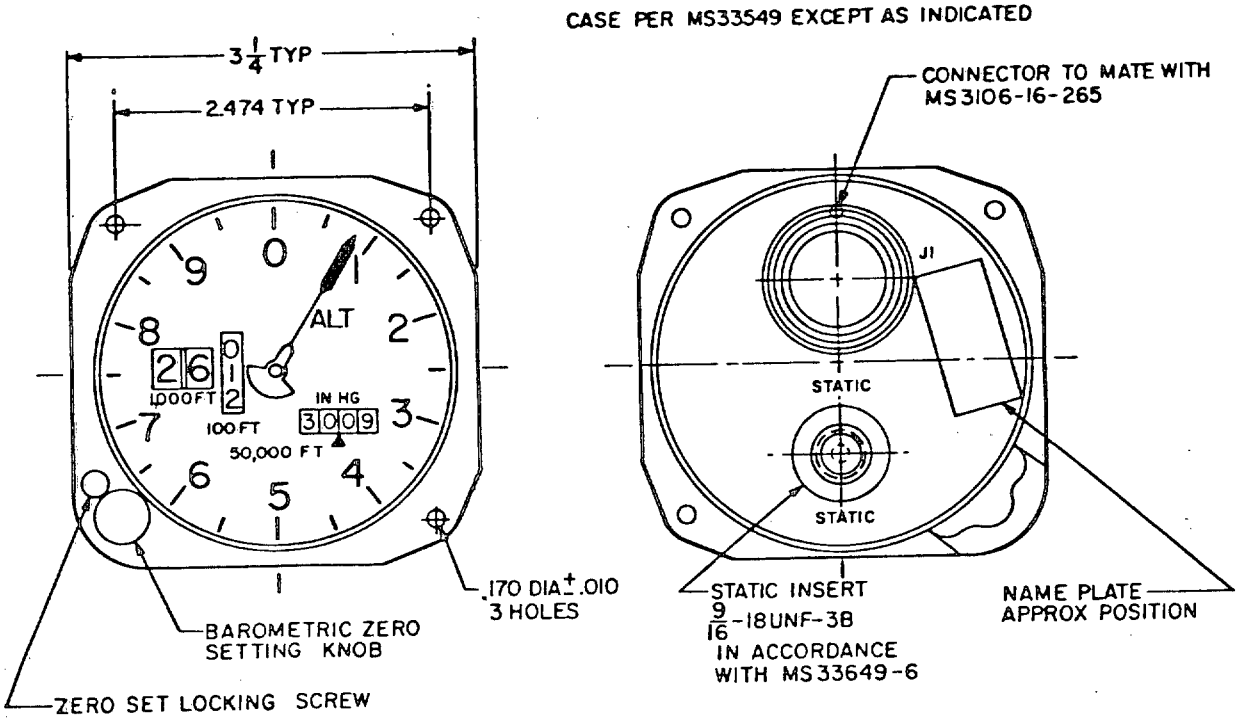
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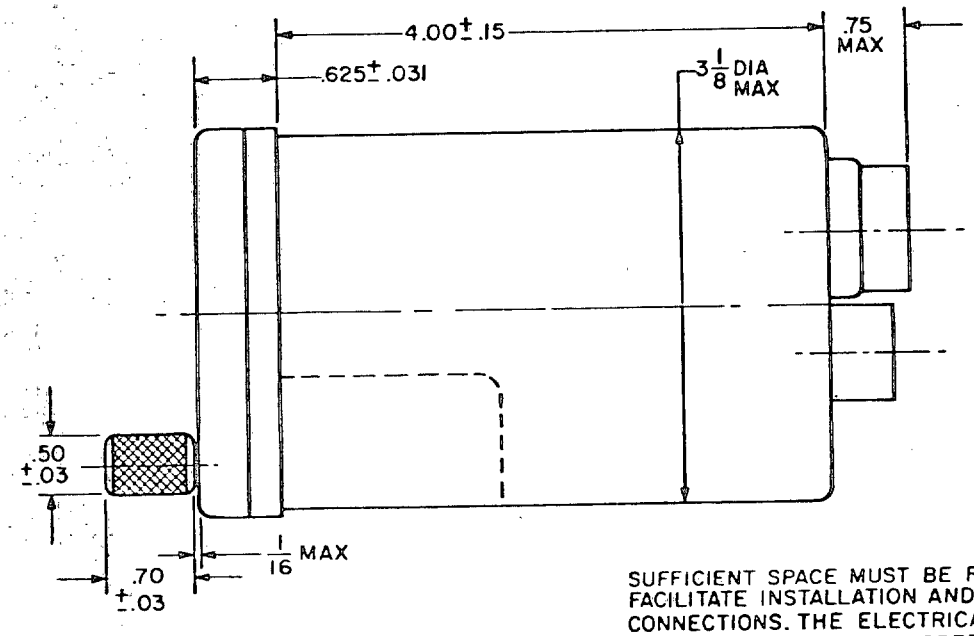
TABLE IV

HYSTERESIS

In. Hg. Test Point	Standard Altitude	Permissible Variation
1st 11.103	25,000 ft.	60 ft.
2nd 13.750	20,000 ft.	60 ft.
3rd 29.921	0 ft.	30 ft.

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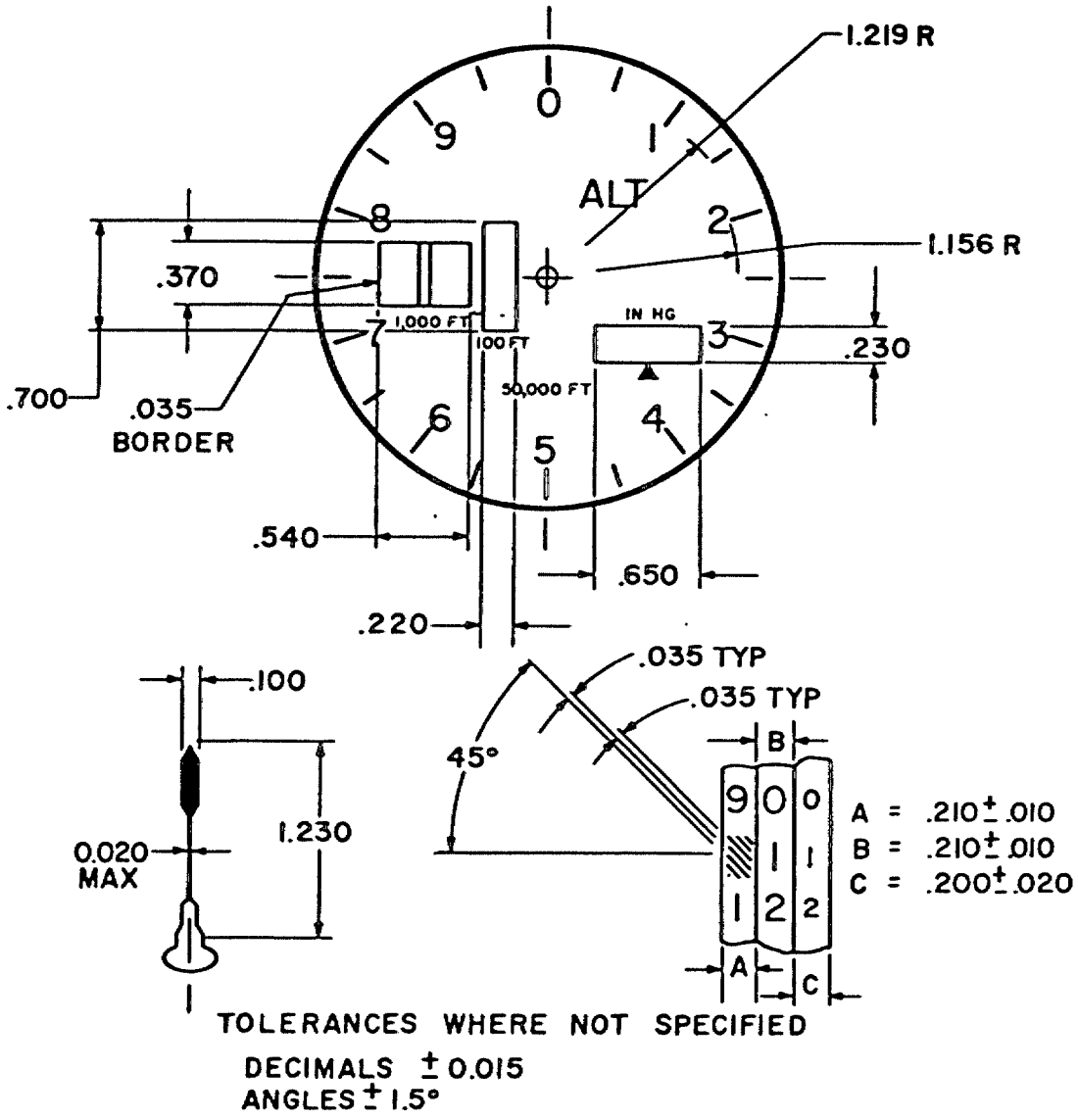
TOLERANCES WHERE NOT SPECIFIED
DECIMALS ± 0.010
FRACTIONS ± 1/64

SUFFICIENT SPACE MUST BE RETAINED TO FACILITATE INSTALLATION AND REMOVAL OF CONNECTIONS. THE ELECTRICAL CONNECTOR SHALL BE LOCATED IN UPPER HALF AND ON VERTICAL ℄. STATIC CONNECTION SHALL BE LOCATED IN LOWER HALF AREA AS DICTATED BY INTERNAL DESIGN. THE SUMP MAY EXTEND THE LENGTH OF THE CASE.

NOTE: DIAL MARKING AAU-31/A-1 OR AAU-31/A-2 AS APPLICABLE.

FIGURE 1.

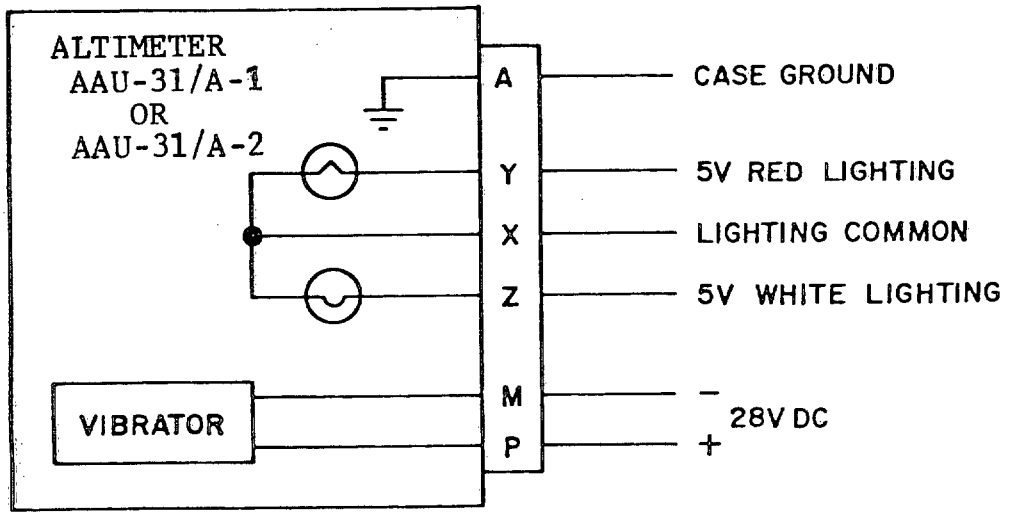
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NOTE: DIAL MARKING AAU-31/A-1 OR AAU-31/A-2 AS APPLICABLE.

FIGURE 2.

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NOTE: TYPE AAU-31/A-1 ALTIMETER INCORPORATES "BLUE FILTER" WHITE LIGHTING.

TYPE AAU-31/A-2 ALTIMETER INCORPORATES "INSTRUMENT AND PANEL" WHITE LIGHTING.

FIGURE 3. Electrical wiring diagram.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-81851B	Altimeter AAU-31/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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